



Depreciation Calculator

Comparison of Straight-Line and Diminishing Balance Methods

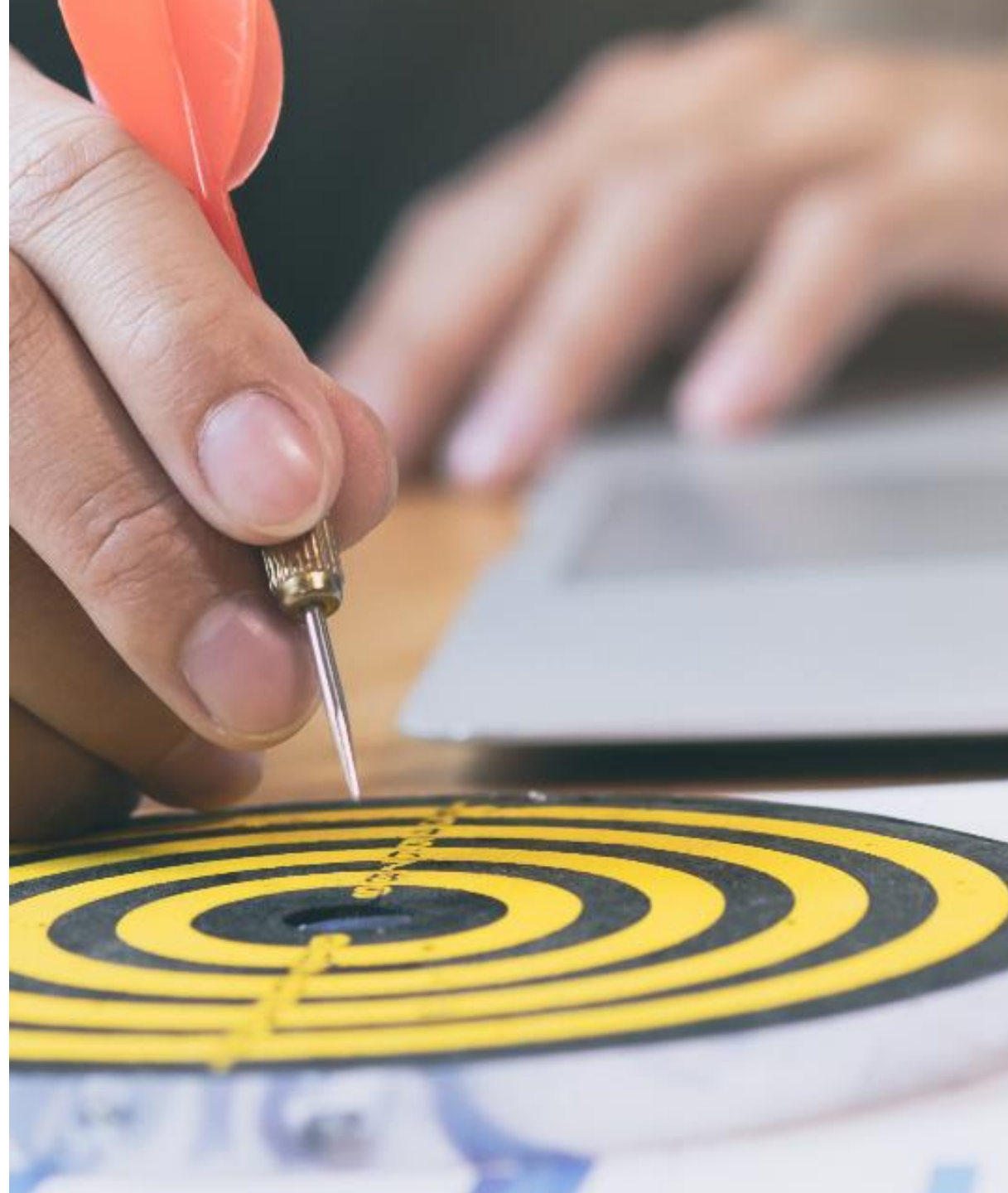


Introduction

- **Depreciation** is systematic reduction of asset value over time.
- Important for financial reporting, asset valuation, and tax deduction.
- Aim: Compare **Straight-Line Method** vs **Diminishing Balance Method** using Excel & answer given problem sets.

Objectives

- Calculate depreciation using Straight-Line & Diminishing Balance methods
- Prepare 10-year schedules.
- Solve Basic / Intermediate / Advanced questions.
- Compare both methods with graphs & insights.





Methodology

- **Inputs:**

- Asset Cost, Additional Costs, Asset Price, Scrap Value, Life Span, Rate of Depreciation (WDV), and Balance Amount.

- **Formulas Used:**

- $\text{Asset Price} = \text{Asset Cost} + \text{Additional Costs}$
- $\text{Annual Dep (SLM)} = (\text{Asset Price} - \text{Scrap Value}) \div \text{Life Span}$
- $\text{Total Dep (SLM)} = \text{Annual Dep} \times \text{Life Span}$
- $\text{Dep (DBM)} = \text{Opening Book Value} \times \text{Rate of Depreciation}$

- **Conceptual Basis:**

- **SLM:** Spreads cost evenly, assuming constant utility over life.
- **Diminishing Balance Method** - Applies higher depreciation in early years, reflecting faster value loss.
- **Validation:** In both methods, $\text{Total Dep} \approx \text{Asset Price} - \text{Scrap Value}$; $\text{Closing Book Value} = \text{Scrap Value}$.

Basic Problem Statements

Straight Line Method (SLM):

Q1. Calculate the Asset Price

- Formula: $\text{Asset Price} = \text{Asset Cost} + \text{Additional Costs}$
- Answer: \$5,00,000.00

Q2. What is the depreciation as per straight-line method?

- Formula: $(\text{Asset Price} - \text{Scrap Value}) \div \text{Life Span}$
- Excel: =SLN(AssetPrice, Scrap, Life)
- Answer: \$45,000.00

Q3. What is the depreciation percentage for the straight-line method?

- Formula: $(\text{Annual Depreciation} \div \text{Asset Price}) \times 100$
- Answer: 9.00%

Basic Problem Statements

Q4. What is the total depreciation for its life span?

- Formula: $\text{Annual Depreciation} \times \text{Life Span}$
- Should equal Asset Price – Scrap Value
- Answer: **\$4,50,000.00**

Q5. Find the depreciated book value after its life span

- Formula: $\text{Asset Price} - \text{Total Depreciation}$
- Should equal Scrap Value
- Answer: **\$50,000.00**

Q6. What is the Balance amount?

- Balance Amount = Book Value after Life Span
- Answer: **\$0**

Basic Problem Statements

Diminishing Balance Method (WDV):

Q1. Calculate the Asset Price

- Formula: Asset Price = Asset Cost + Additional Costs
- Answer: \$5,00,000.00

Q2. Find the rate of depreciation as per diminishing balance method

- Excel Formula: =IF(D20="", "", 1-(D21/D20)^(1/D22))
- Answer: 20.57%

Intermediate Problem Statements

Q1. Book Value for Year 1 & Depreciation for Year 1

•Answer: Book Value (Y1) = \$5,00,000.00 & Depreciation (Y1) = \$1,02,835.88

Q2. Book Value for Year 2 & Depreciation for Year 2

•Answer: Book Value (Y2) = \$3,97,164.12 & Depreciation (Y2) = \$81,685.45

Q3. Book Value for Year 3 & Depreciation for Year 3

•Answer: Book Value (Y3) = \$3,15,478.67 & Depreciation (Y3) = \$64,885.06

Q4. Extend to 10 years

Year	Year on Year Depreciation Amount	Book Value
1	\$1,02,835.88	\$5,00,000.00
2	\$81,685.45	\$3,97,164.12
3	\$64,885.06	\$3,15,478.67
4	\$51,540.03	\$2,50,593.62
5	\$40,939.70	\$1,99,053.59
6	\$32,519.56	\$1,58,113.88
7	\$25,831.21	\$1,25,594.32
8	\$20,518.46	\$99,763.12
9	\$16,298.39	\$79,244.66
10	\$12,946.27	\$62,946.27
		\$50,000.00

Advanced Problem Statements

Q1) Annual depreciation (Straight-Line)

Answer = \$45,000.00

Formula used: $=(\text{AssetPrice} - \text{Scrap}) / \text{Life}$

Q2) Total depreciation for whole life (Straight-Line)

Answer: = \$4,50,000.00

Formula used: $=(\text{AssetPrice} - \text{Scrap})$

Q3) Book value after life (Straight-Line)

Answer: = \$50,000.00

Formula used: $= \text{AssetPrice} - ((\text{AssetPrice} - \text{Scrap}) / \text{Life}) * \text{Life}$ [It should be equal to scrap value]

Q4) Rate of depreciation per year (Diminishing Balance)

Answer = 20.57%

Formula used: $= 1 - (\text{Scrap} / \text{AssetPrice}) ^ (1 / \text{Life})$

Advanced Problem Statements

Q5) Depreciation in Year 2 (Diminishing Balance)

Answer = **\$81,685.45**

Formula = $[\text{Asset price} - \text{Dep (Y1)}] * \text{diminishing_rate}$

Q6) Book value at end of Year 4 (Diminishing Balance)

Answer = **\$2,50,593.62**

Formula used: = Book value (Y3) – Depreciation Amount (Y3)

Q7) Total depreciation for whole life (Diminishing Balance)

Answer = **\$4,37,053.73**

Formula used: = Book Value (Y1) – Book Value (Y10)

Q8) Book value after life (Diminishing Balance)

Answer = **\$50,000.00**

Formula used = Scrap Value

Advanced Problem Statements

Q9) Compare total depreciation from Straight Line & diminishing Method. Which is higher?

Answers to show in sheet:

- Total Depreciation (SLM) = \$4,50,000.00
- Total Depreciation (DBM) = \$4,50,000.00
- Total Depreciation amount is **same** for both the methods.

Comparison & Analysis

- Straight-line Method

Year	Opening Book Value	Depreciated Value	Closing Book Value
1	\$5,00,000.00	\$45,000.00	\$4,55,000.00
2	\$4,55,000.00	\$45,000.00	\$4,10,000.00
3	\$4,10,000.00	\$45,000.00	\$3,65,000.00
4	\$3,65,000.00	\$45,000.00	\$3,20,000.00
5	\$3,20,000.00	\$45,000.00	\$2,75,000.00
6	\$2,75,000.00	\$45,000.00	\$2,30,000.00
7	\$2,30,000.00	\$45,000.00	\$1,85,000.00
8	\$1,85,000.00	\$45,000.00	\$1,40,000.00
9	\$1,40,000.00	\$45,000.00	\$95,000.00
10	\$95,000.00	\$45,000.00	\$50,000.00

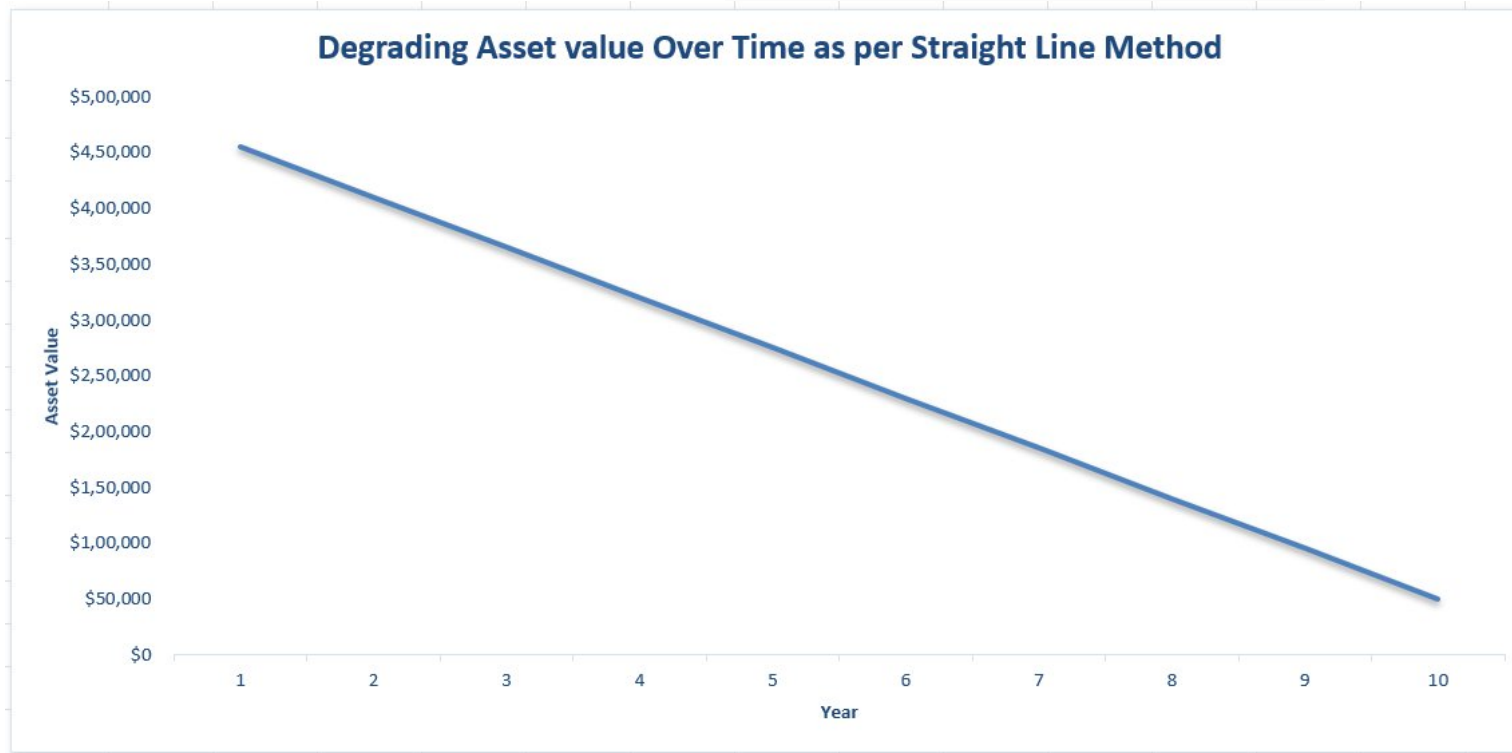
Comparison & Analysis

- Diminishing Balance Method

Year	Year on Year Depreciation Amount	Book Value
1	\$1,02,835.88	\$5,00,000.00
2	\$81,685.45	\$3,97,164.12
3	\$64,885.06	\$3,15,478.67
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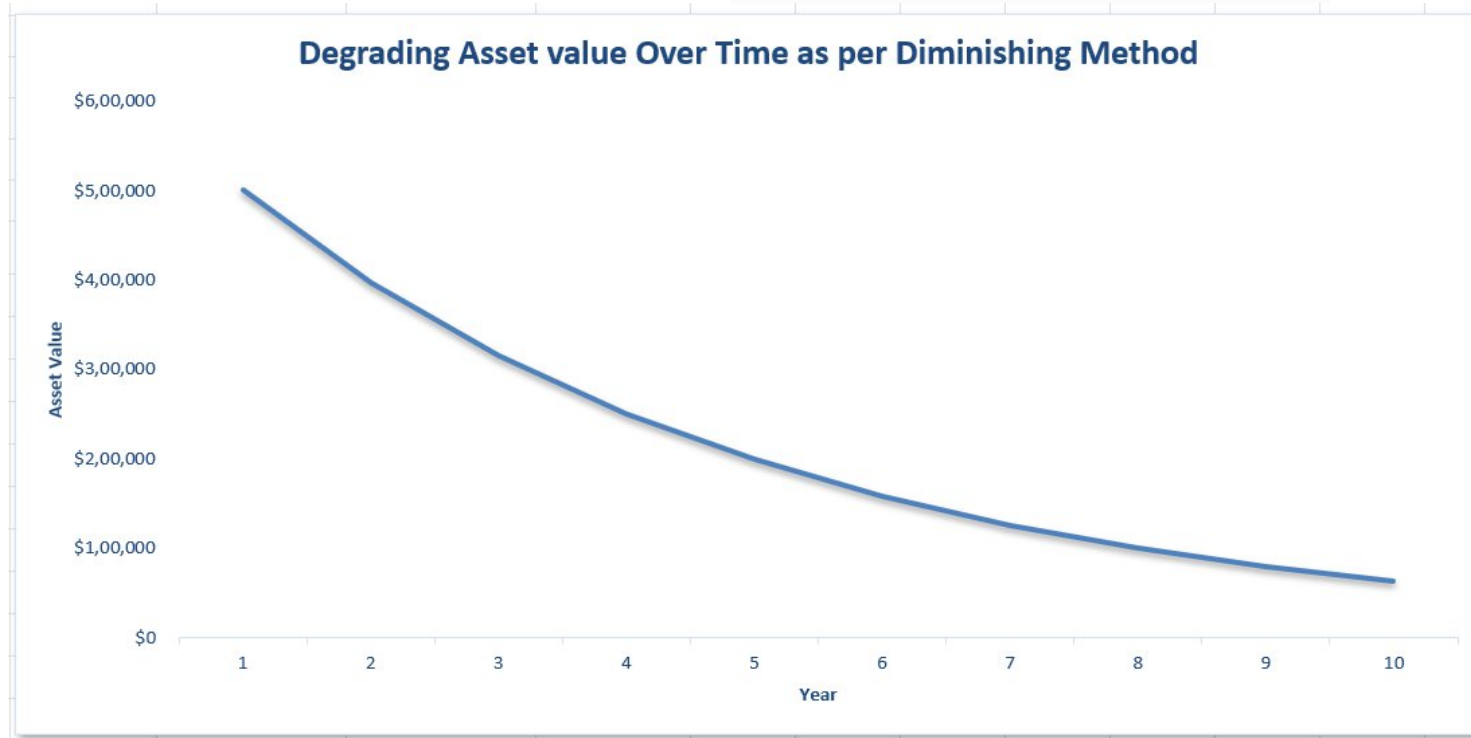
Book/Asset Value vs. Year

- Straight-line Method



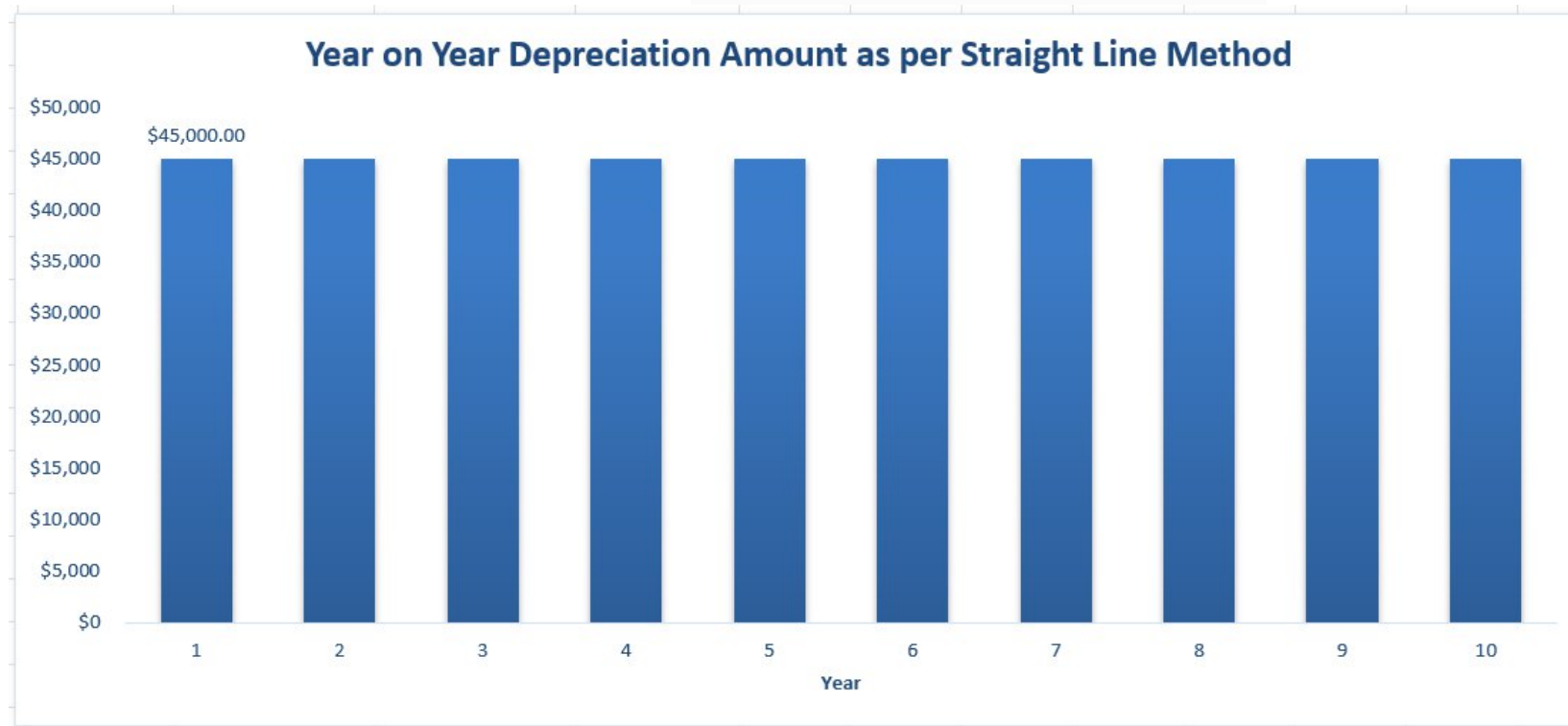
Book/Asset Value vs. Year

- Diminishing Balance Method



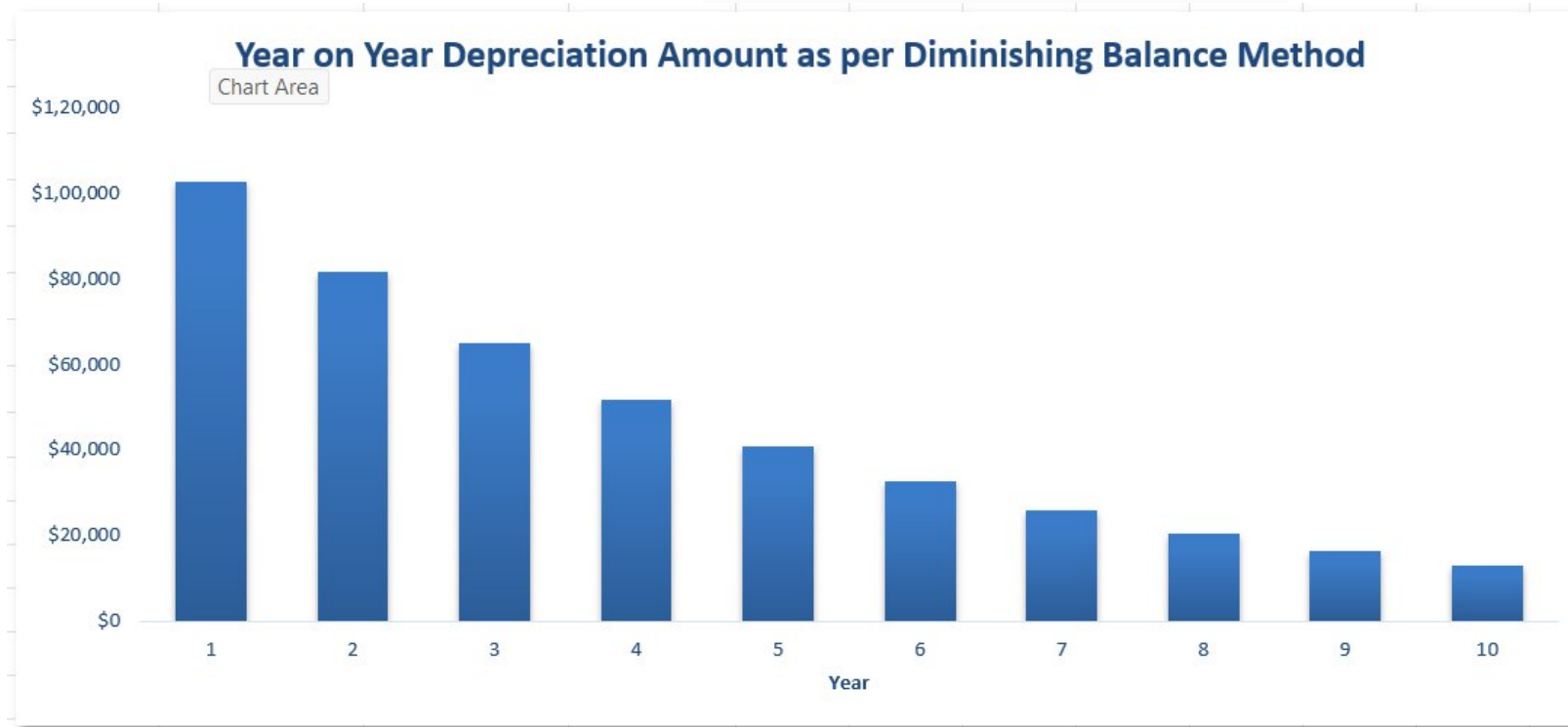
Yearly Depreciation Amount

- Straight-line Method



Yearly Depreciation Amount

- Diminishing Balance Method



Comparison Insights

- **Straight-Line Method (SLM):**

Depreciation expense remains **constant every year**, which simplifies planning and reporting. Ensures **stability in profit & loss statements**, as expenses are evenly distributed. Best suited for assets like **buildings, furniture, or fixtures** that provide a **uniform level of benefit** over their useful life.

- **Diminishing Balance Method:**

Depreciation is **higher in the initial years** and gradually reduces over time. Mirrors the real-world usage pattern of many assets (e.g., machinery, vehicles, technology) that lose efficiency or value quickly in the beginning. Provides a **tax advantage in early years**, since higher depreciation lowers taxable profit initially.

Conclusion

- Both **Straight-Line Method (SLM)** and **Diminishing Balance Method** depreciate the asset by the same total amount (**Asset Price – Scrap Value**) over its useful life.
- The key difference lies in **timing**:
 - **SLM** spreads depreciation evenly, ensuring stability in reporting and budgeting.
 - **Diminishing Balance Method** charges more in early years, reflecting faster wear and providing early tax benefits.
- The **choice of method** depends on the asset's nature:
 - **SLM** suits assets with steady utility (e.g., buildings, furniture).
 - **Diminishing Balance Method** fits assets that lose value quickly (e.g., machinery, vehicles, technology).

Thank you